

- Description

This module manages a smart coffee machine system, ensuring that:

1. The machine continuously monitors the current brew temperature and coffee strength.
2. If the current temperature deviates from the target temperature, the system automatically adjusts the brew time.
3. The coffee strength is also updated based on user input, with alerts if it is too weak or too strong.

This system helps optimize coffee extraction by ensuring ideal brewing conditions, thereby enhancing flavor and consistency.

- Specification

module SmartCoffeeMachineControl

var currentTemperature, targetTemperature, brewTime, coffeeStrength: real

process updateBrewConditions(tempChange: real, strengthChange: real) returns (brewTime: real, coffeeStrength: real)

functional scenario form:

currentTemperature < targetTemperature && brewTime = brewTime * 1.1

currentTemperature > targetTemperature && brewTime = brewTime * 0.9,

output "Updated Conditions: Temperature = " + currentTemperature + "°C, Coffee Strength = " + coffeeStrength + ", Brew Time = " + brewTime + " minutes"

end process

end module

- Java Code

```
package code;
class c73_SmartCoffeeMachine {
    private double targetTemperature;
    private double currentTemperature;
    private double brewTime;
    private double coffeeStrength;
    public c73_SmartCoffeeMachine(double targetTemperature, double initialTemperature, double baseBrewTime, double defaultStrength) {
        this.targetTemperature = targetTemperature;
        this.currentTemperature = initialTemperature;
        this.brewTime = baseBrewTime;
        this.coffeeStrength = defaultStrength;
    }
    public void updateBrewConditions(double tempChange, double strengthChange) {
        currentTemperature += tempChange;
        coffeeStrength += strengthChange;
        adjustBrewTime();
    }
}
```

```

        System.out.printf("Current Temperature: %.2f°C, Coffee Strength
: %.2f, Brew Time: %.2f minutes\n",
                           currentTemperature, coffeeStrength, brewTime)
;
        if (currentTemperature < targetTemperature - 5) {
            System.out.println("ALERT: Temperature too low, consider pr
e-heating.");
        } else if (currentTemperature > targetTemperature + 5) {
            System.out.println("ALERT: Temperature too high, risk of ov
er-extraction.");
        }
        if (coffeeStrength < 3) {
            System.out.println("ALERT: Coffee too weak, consider adjust
ing grind size or amount.");
        } else if (coffeeStrength > 8) {
            System.out.println("ALERT: Coffee too strong, may be over-
extracted.");
        }
    }
    private void adjustBrewTime() {
        if (currentTemperature < targetTemperature) {
            brewTime = brewTime * 1.1;
        } else {
            brewTime = brewTime * 0.9;
        }
    }
    public double getCycleTime() {
        return brewTime;
    }
    public double getCoffeeStrength() {
        return coffeeStrength;
    }
    public static void main(String[] args) {
        c73_SmartCoffeeMachine machine = new c73_SmartCoffeeMachine(90.0,
80.0, 5.0, 5.0);
        // testcase-VT:
        //     machine.updateBrewConditions(3.0, 2.0);
        //     machine.updateBrewConditions(10.0, -1.0);
        //     machine.updateBrewConditions(-15.0, 0.0);
        //     machine.updateBrewConditions(15.0, -10.0);
        //     machine.updateBrewConditions(9.0, 15.0);
        //     machine.updateBrewConditions(-100.0, 15.0);

        // testcase-FT:

```

```
//      machine.updateBrewConditions(7.325806634504615E300, 0.0);
//      machine.updateBrewConditions(-1.7976930616042493E308, 0.0);
//      machine.updateBrewConditions(-9.058687296395083E307, -
1.7976931348623157E308);
//      machine.updateBrewConditions(4.915567166162387E307, 0.0);
//      machine.updateBrewConditions(-4.9E-324, 0.0);
//      machine.updateBrewConditions(-4.9E-324, -
9.058687296395083E307);

// testcase-Z3:
//      machine.updateBrewConditions(3.0, 2.0);
//      machine.updateBrewConditions(10.0, -1.0);
//      machine.updateBrewConditions(-15.0, 0.0);
//      machine.updateBrewConditions(15.0, -10.0);
//      machine.updateBrewConditions(9.0, 15.0);
//      machine.updateBrewConditions(-100.0, 15.0);

// testcase-UVT:
machine.updateBrewConditions(3.0, 2.0);
machine.updateBrewConditions(10.0, -1.0);
machine.updateBrewConditions(-15.0, 0.0);
machine.updateBrewConditions(15.0, -10.0);
machine.updateBrewConditions(9.0, 15.0);
machine.updateBrewConditions(-100.0, 15.0);
}
}
```

- Coverage report

1. Test case number: 6
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.2 %	10	832	842
▼ src	1.2 %	10	832	842
▼ code	1.2 %	10	832	842
▼ c73_SmartCoffeeMachine.java	100.0 %	10	0	10
▼ c73_SmartCoffeeMachine	100.0 %	10	0	10
main(String[])	100.0 %	0	0	0
c73_SmartCoffeeMachine(double, double)	100.0 %	0	0	0
adjustBrewTime()	100.0 %	2	0	2
updateBrewConditions(double, double)	100.0 %	8	0	8

3. Vibration testing instruction coverage: 96.1%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.7 %	147	20,817	20,964
▼ src	0.7 %	147	20,817	20,964
▼ code	0.7 %	147	20,817	20,964
▼ c73_SmartCoffeeMachine.java	96.1 %	147	6	153
▼ c73_SmartCoffeeMachine	96.1 %	147	6	153
main(String[])	100.0 %	33	0	33
c73_SmartCoffeeMachine(double, double)	100.0 %	15	0	15
adjustBrewTime()	100.0 %	20	0	20
updateBrewConditions(double, double)	100.0 %	79	0	79

4. Fuzz testing branch coverage: 80%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c73_SmartCoffeeMachine.java	80.0 %	8	2	10
▼ c73_SmartCoffeeMachine	80.0 %	8	2	10
main(String[])		0	0	0
c73_SmartCoffeeMachine(double)		0	0	0
adjustBrewTime()	100.0 %	2	0	2
updateBrewConditions(double, int)	75.0 %	6	2	8

5. Fuzz testing instruction coverage: 94.1%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.7 %	144	20,820	20,964
▼ src	0.7 %	144	20,820	20,964
▼ code	0.7 %	144	20,820	20,964
▼ c73_SmartCoffeeMachine.java	94.1 %	144	9	153
▼ c73_SmartCoffeeMachine	94.1 %	144	9	153
main(String[])	100.0 %	33	0	33
c73_SmartCoffeeMachine(double)	100.0 %	15	0	15
adjustBrewTime()	100.0 %	20	0	20
updateBrewConditions(double, int)	96.2 %	76	3	79

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.2 %	10	832	842
▼ src	1.2 %	10	832	842
▼ code	1.2 %	10	832	842
▼ c73_SmartCoffeeMachine.java	100.0 %	10	0	10
▼ c73_SmartCoffeeMachine	100.0 %	10	0	10
main(String[])		0	0	0
c73_SmartCoffeeMachine(double)		0	0	0
adjustBrewTime()	100.0 %	2	0	2
updateBrewConditions(double, int)	100.0 %	8	0	8

7. Z3 instruction coverage: 96.1%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.7 %	147	20,817	20,964
▼ src	0.7 %	147	20,817	20,964
▼ code	0.7 %	147	20,817	20,964
▼ c73_SmartCoffeeMachine.java	96.1 %	147	6	153
▼ c73_SmartCoffeeMachine	96.1 %	147	6	153
main(String[])	100.0 %	33	0	33
c73_SmartCoffeeMachine(double)	100.0 %	15	0	15
adjustBrewTime()	100.0 %	20	0	20
updateBrewConditions(double, int)	100.0 %	79	0	79

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.2 %	10	832	842
▼ src	1.2 %	10	832	842
▼ code	1.2 %	10	832	842
▼ c73_SmartCoffeeMachine.java	100.0 %	10	0	10
▼ c73_SmartCoffeeMachine	100.0 %	10	0	10
main(String[])		0	0	0
c73_SmartCoffeeMachine(double)		0	0	0
adjustBrewTime()	100.0 %	2	0	2
updateBrewConditions(double, int)	100.0 %	8	0	8

9. Unimproved V-method instruction coverage: 96.1%

Element		Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼  VibrationTestingTool		0.7 %	147	20,817	20,964
▼  src		0.7 %	147	20,817	20,964
▼  code		0.7 %	147	20,817	20,964
▼  c73_SmartCoffeeMachine.java		96.1 %	147	6	153
▼  c73_SmartCoffeeMachine		96.1 %	147	6	153
 main(String[])		100.0 %	33	0	33
 c73_SmartCoffeeMachine(double)		100.0 %	15	0	15
 adjustBrewTime()		100.0 %	20	0	20
 updateBrewConditions(double, double)		100.0 %	79	0	79